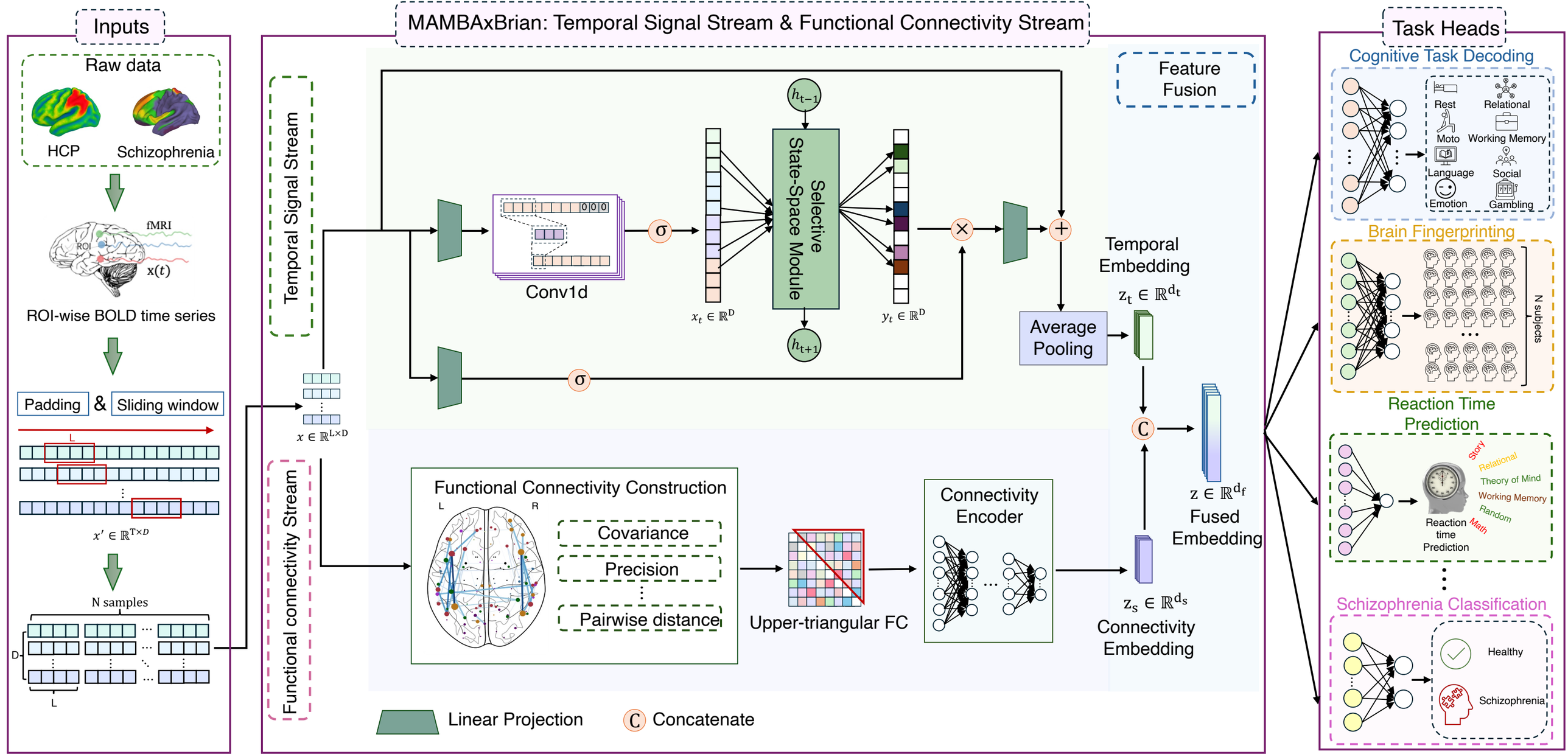

Inputs
Raw data
HCP
Schizophrenia
fMRI
ROI
x(t)
ROI-wise BOLD time series
Padding
&
Sliding window
L
$x' \in \mathbb{R}^{T \times D}$
N samples
D
L
MAMBAxBrian: Temporal Signal Stream & Functional Connectivity Stream
Temporal Signal Stream
Conv1d
σ
h_{t-1}
Selective State-Space Module
h_{t+1}
$x_t \in \mathbb{R}^D$
$y_t \in \mathbb{R}^D$
Feature Fusion
Average Pooling
Temporal Embedding
$z_t \in \mathbb{R}^{d_t}$
$x \in \mathbb{R}^{L \times D}$
Functional connectivity Stream
Functional Connectivity Construction
L
R
Covariance
Precision
Pairwise distance
Upper-triangular FC
Connectivity Encoder
$z_s \in \mathbb{R}^{d_s}$
Connectivity Embedding
C
$z \in \mathbb{R}^{d_f}$
Fused Embedding
Linear Projection
Concatenate
Task Heads
Cognitive Task Decoding
Rest
Relational
Moto
Working Memory
Language
Social
Emotion
Gambling
Brain Fingerprinting
N subjects
Reaction Time Prediction
Story
Relational
Theory of Mind
Working Memory
Random
Math
Reaction time Prediction
Schizophrenia Classification
Healthy
Schizophrenia